AS MATHEMATICS - PURE 1 - DAY 6

Circles

Centre, radius, equations and tangents

YOUR AIM TODAY

Read and build circle equations, then use radius-tangent geometry.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

A circle is every point the same distance from its centre.

- Centre (a,b), radius r: $(x-a)^2+(y-b)^2=r^2$.
- The signs reverse inside brackets: $(x+3)^2$ gives centre $x=-3$.
- Expand only when asked; completed-square form shows the geometry.
- A tangent is perpendicular to the radius at the point of contact.

MEMORY RULE

Centre signs reverse; tangent gradient is the negative reciprocal of radius gradient.

Read a circle

WORKED STEPS

$(x-2)^2+(y+1)^2=25$ has centre $(2,-1)$ and radius 5.

Tangent

WORKED STEPS

Centre $(0,0)$, point $(3,4)$: radius gradient $4/3$, tangent gradient $-3/4$; $y-4=(-3/4)(x-3)$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. State centre and radius of (x+1)^2+(y-4)^2=9.	
2. Write the circle with centre (3,-2), radius 4.	
3. Does (4,3) lie on x^2+y^2=25?	
4. Complete squares: x^2+y^2-6x+4y-12=0.	
5. Find the tangent to x^2+y^2=25 at (3,4).	
6. Find intersections of x^2+y^2=25 with y=4.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	Centre (-1,4), radius 3.
2	(x-3)^2+(y+2)^2=16.
3	Yes: 16+9=25.
4	(x-3)^2+(y+2)^2=25.
5	y-4=(-3/4)(x-3), or 3x+4y=25.
6	x^2+16=25, so x=3 or -3: (3,4), (-3,4).

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.